

Problem 1

- a) Linear speedup refers to an increase in performance with the increase of nb. of cores.
The increase is linear.

For example:

$$\begin{aligned} 1 \text{ core} &\rightarrow \text{time } T \\ 2 \text{ cores} &\rightarrow \text{time } \frac{T}{2} \Rightarrow S = 2 \\ 4 \text{ cores} &\rightarrow \frac{T}{4} \Rightarrow S = 4 \end{aligned}$$

Basically the speedup is proportional with nb. of cores. $\Rightarrow S(n) = n$

- b) Superlinear speedup can be achieved in practice only by some anomalies and caching.
In theory superlinear speedup can't be achieved.

- c) Efficiency shows how much we use from the capabilities of the system.

$$E_n = \frac{S_n}{n} = \frac{\text{speedup}}{\text{nb. cores.}}$$

An efficiency of 0.5 shows that we are using only a half of the system's potential.

5 processors
+ hyperthread
→ something like 6.5 cores

- d) Absolute speedup uses $T_s \rightarrow$ serial time for computation $\Rightarrow S_n = \frac{T_s}{T_p(n)}$
Relative speedup uses $T_p(1) \rightarrow$ using 1 core $\Rightarrow S_n = \frac{T_p(1)}{T_p(n)}$

- e) 5 processors
+ hyperthread
→ something like 6.5 cores.

$$\Rightarrow \text{Max speedup} = 6.5$$

$$\Rightarrow \text{Max efficiency} = 1 \text{ (or 100\%)}$$

- f) Parameters reg and par are fractions of time.

In order to calculate them, we ~~divide~~ delimit the serial and parallelizable parts and run the problem

with 1 core.

$$T(1) = t_s + t_p$$

After that we measure the time it takes to solve the serial part, t_s

$$\Rightarrow \text{reg} = \frac{t_s}{T(1)}$$

$$\Rightarrow \text{par} = 1 - \text{reg}$$

Example:

$$\text{reg} = 0.41$$

$$\text{par} = 0.59$$

Problem 1 - continued

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g) Assumptions:

- problem has constant size
- all cores are equal in all aspects
- parallel overhead is minimum
- memory is uniform and problem fits into memory